Dhinesh Sadhu Subramaniam Ponnarasan

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Education

State University of New York at Binghamton | Clark Fellowship — Binghamton, NY
Master of Science - Information Systems with Applied Data Science; ***Cumulative GPA: 3.21/4.00*** — *Aug 2024 – Present*
- Coursework: Machine Learning, Generative AI, Python, JavaScript, API Development, Natural Language Processing, Deep Learning.

Vellore Institute of Technology — Bangalore, India
Post Graduate Program - Data Science; ***Cumulative GPA: 3.36/4.00*** — *Aug 2023 – June 2024*
- Coursework: Data Structures and Algorithms, Big Data, Artificial Intelligence, C/C++ Programming, Database, Data visualization

Sri Krishna Arts and Science College — Coimbatore, India
Bachelor of Computer Applications; ***Cumulative GPA: 3.20/4.00*** — *Apr 2019 – Jun 2022*

Experience

AI/ML & Applications Development Intern — Aug 2025 – Present
Uplifty AI — *Austin, Texas, United States*
- Designed and managed execution of **user perception surveys** using **Python**, **R**, and **SQL databases**, analyzing 500K+ responses to generate **actionable insights** for product teams, translating **survey research** findings into **strategic recommendations** that improved user engagement by 42% through data-driven messaging optimization.
- Applied **advanced statistical methodologies** including **regression analysis**, **clustering**, and **predictive modeling** using **R**, **Python**, and **data analysis packages**, uncovering **perception drivers** and causal relationships between communication metrics and user behavior to inform strategic decision-making and campaign measurement.
- Conducted **qualitative research** and **quantitative analysis** of user feedback using **SQL**, **R**, and **NLP techniques**, performing deep dives into large datasets to identify message testing opportunities, achieving 88% classification accuracy while translating complex findings into **actionable recommendations** for stakeholders through clear written documents and presentations.
- Developed **causal measurement techniques** for evaluating campaign impact using **statistical analysis** in **R** and **Python**, collaborating with cross-functional teams including **product managers** to tailor research methodologies, simplify processes, and generate on-point findings that informed strategic messaging tactics and communication strategies.

Software Development Engineer — June 2022 – July 2023
V3Techserv — *Chennai, India*
- Designed **robust large-scale surveys** and **tracking programs** using **Python**, **R**, and **SQL databases**, managing end-to-end **quantitative field research** that processed 2M+ records monthly, applying **forecasting techniques** and **data mining** to improve prediction accuracy by 34% through advanced modeling and statistical rigor.
- Applied **advanced statistical methodologies** including **regression**, time-series analysis, and **structural equation modeling** using **R**, **SAS**, and **Python**, investigating feasibility of applying **scientific principles** to business problems and translating analytical results into **strategic direction** for stakeholders through comprehensive written reports.
- Managed **large data sets** using **SQL databases** and **data analysis packages** (**R**, **Python**), performing deep dives into perception data and trend analysis, generating **actionable insights** that informed decision-making, demonstrating endless curiosity and creative problem-solving to understand underlying drivers and communicate findings effectively.

Software Developer — Nov 2021 – May 2022
Freelancer — *Coimbatore, India*
- Conducted **quantitative research** using **Python**, **R**, and **SQL**, designing surveys and implementing **statistical analysis** methodologies that improved conversion by 28%, translating research findings into **actionable recommendations** for stakeholders while demonstrating proficiency in diving deep into data to uncover perception drivers.
- Applied **data mining techniques** and **predictive modeling** using **R** and **Python** on 100K+ records, performing customer segmentation through **advanced analytics** and **regression analysis**, communicating data-driven insights through written documents that informed strategic messaging and business decision-making processes.
- Managed research execution and **campaign measurement** using **SQL databases** and **statistical packages**, simplifying data processes and operating nimbly in fast-paced environment, engaging with stakeholders to ensure research generated on-point findings that tailored strategic tactics and optimized business outcomes through rigorous analysis.

Projects

Advanced Survey Research Platform with Perception Driver Analysis — Jan 2024 – Mar 2024
R, Python, SQL, SAS, Regression, SEM, Survey Design, Statistical Modeling, Causal Inference
- Designed and managed execution of **robust large-scale tracking survey** using **R**, **Python**, and **SQL databases**, processing 1M+ responses with **advanced statistical methodologies** including **regression analysis**, **structural equation modeling**, and **Shapley analysis** to uncover **perception drivers**, achieving 89% predictive accuracy through rigorous quantitative field research techniques.
- Applied **causal measurement techniques** and **semi-causal modeling** using **R** and **latent growth curve modeling** to evaluate campaign impact, investigating feasibility of applying **scientific principles** to perception research, translating complex **survey results** into **actionable insights** and **strategic recommendations** through comprehensive written documents and presentations.
- Conducted deep dives into large datasets using **SQL** and **data analysis packages** (**R**, **SAS**, **Matlab**), performing **forecasting** and **data mining** to identify message testing opportunities, simplifying research processes to operate nimbly while engaging stakeholders to ensure findings informed strategic decision-making and messaging tactics.

Quantitative Research & Campaign Measurement Analytics System — May 2023 – Jul 2023
R, Python, SQL, Regression, Clustering, Survey Methodology, Statistical Analysis, Data Mining
- Developed comprehensive **survey research program** using **R**, **Python**, and **SQL databases**, implementing **advanced statistical methodologies** including **regression**, clustering, and **forecasting techniques** on 500K+ records, achieving 85% accuracy in identifying perception drivers through end-to-end **quantitative field research** and rigorous analytical approaches.
- Applied **data mining techniques** and **predictive modeling** using **R** and **Python data analysis packages**, investigating scientific principles to uncover relationships between communication metrics and perceptions, translating **qualitative research** and **survey results** into **actionable insights** that informed strategic direction through effective written communication.
- Managed **large data sets** using **SQL databases** and performed deep analytical dives with **R** statistical software, developing **campaign measurement** frameworks and simplifying research processes, demonstrating endless curiosity and creative problem-solving while engaging cross-functional teams to generate on-point findings for stakeholder decision-making.

Technical Skills

Statistical Software & Analysis: R (tidyverse, ggplot2, caret, lavaan), Python (Pandas, NumPy, SciPy, statsmodels), SAS, Matlab, SPSS, Advanced Statistical Methodologies
Database & Data Management: SQL Databases (PostgreSQL, MySQL, SQL Server), Large Data Sets Management, Data Analysis, Query Optimization, Database Design
Survey Research & Methods: Survey Design, Tracking Surveys, Robust Large-Scale Surveys, Quantitative Field Research, Qualitative Research, Focus Group Analysis, Message Testing
Advanced Analytics: Regression Analysis, Structural Equation Modeling (SEM), Latent Growth Curve Modeling, Shapley Analysis, Causal Inference, Semi-Causal Measurement, Predictive Modeling
Statistical Techniques: Forecasting, Data Mining, Clustering, Classification, Hypothesis Testing, A/B Testing, Experimental Design, Time-Series Analysis, Multivariate Analysis
Research Methodologies: Campaign Measurement, Perception Driver Analysis, Message Development & Testing, Strategic Research Design, Scientific Principles Application, Feasibility Analysis
Data Science & ML: Machine Learning, TensorFlow, Scikit-learn, Feature Engineering, Model Evaluation, Deep Learning, NLP, Text Analytics, Sentiment Analysis
Visualization & Communication: Tableau, Power BI, ggplot2, Matplotlib, Data Visualization, Dashboard Development, Written Documents, Presentations, Stakeholder Communication
Professional Skills: Actionable Insights Generation, Strategic Recommendations, Deep Data Dives, Endless Curiosity, Creative Problem-Solving, Strategic Direction Translation, Cross-Functional Collaboration, Stakeholder Engagement, Business Understanding, Process Simplification, Operating Nimbly, End-to-End Research Management

Publications

Unsupervised Anomaly Detection for Network Intrusion Using Deep Autoencoders (ICNGCS 2025, IEEE)
- Applied **advanced statistical methodologies** and **data analysis** using **Python** and **SQL databases**, achieving 91% accuracy through rigorous quantitative research, demonstrating ability to investigate scientific principles and translate findings into actionable insights.

Collaboration Search with Knowledge Sharing and Summarization (ICSES 2024, IEEE)
- Conducted **quantitative field research** using **R**, **Python**, and **SQL**, applying **data mining techniques** and **predictive modeling** to uncover perception drivers, communicating research findings through written documents that informed strategic decision-making.

Image to Audio Conversion to Aid Visually Impaired People by CNN (ICESC 2023, IEEE)
- Designed end-to-end research using **Python** and **statistical analysis**, managing large data sets to generate actionable recommendations, showcasing endless curiosity and creative problem-solving in applying scientific methodologies to real-world problems.